BS EN 50129:2018



BSI Standards Publication

Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signalling

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EUROPEAN STANDARD

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English Version

Railway applications - Communication, signalling and processing systems - Safety related electronic systems for signalling

Applications ferroviaires - Systèmes de signalisation, de télécommunications et de traitement - Systèmes électroniques de sécurité pour la signalisation

Bahnanwendungen - Telekommunikationstechnik, Signaltechnik und Datenverarbeitungssysteme - Sicherheitsrelevante elektronische Systeme für Signaltechnik

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3 Terms, definitions and abbreviations

3.1 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

3.1.1
accident
unintended event or series of events that results in death, injury, loss of a system or service, or environmental damage

[SOURCE: IEC 60050-821:2017, 821-12-02]

3.1.2
basic integrity
integrity attribute for safety-related functions with a TFFR higher than (less demanding) 10^{-5} h^{-1} or for non-safety-related functions

Note 1 to entry: In this document Basic Integrity requirements relate only to safety-related functions. If a non-safety-related function has been given basic-integrity requirements on the basis of the process described in EN 50126-2:2017, no additional requirements are defined in this document.

[SOURCE: EN 50126-1:2017, 3.7, modified – The note 1 to entry has been added.]

3.1.3
causal analysis
analysis of the reasons how and why a particular hazard can come into existence

[SOURCE: IEC 60050-821:2017, 821-12-07]

3.1.4
common-cause failures
failures of multiple items, which would otherwise be considered independent of one another, resulting from a single cause

[SOURCE: IEC 60050-192:2015, 192-03-18]

3.1.5
configuration
structuring and interconnection of the hardware and software of a system for its intended application

[SOURCE: IEC 60050-821:2017, 821-12-12]

3.1.6
consequence analysis
analysis of events which are likely to happen after a hazard has occurred

[SOURCE: IEC 60050-821:2017, 821-12-14]

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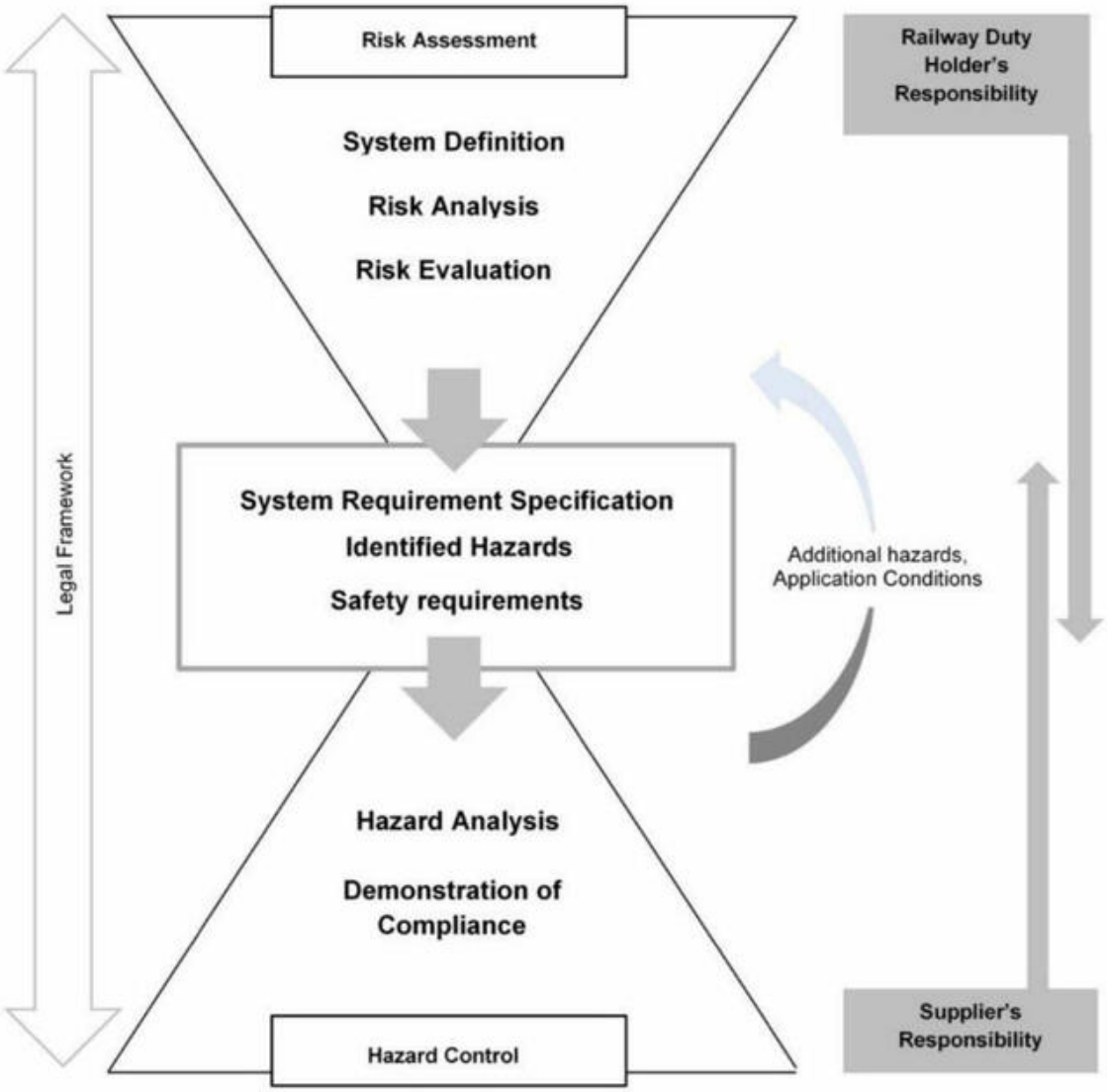


Figure A.2 — Global process overview

A.4.2 Risk Assessment

A.4.2.1 General

The aim of the Risk Assessment is to refine the product hazards, identify their causes and consequences, evaluate the risk, and define the mitigation strategy in compliance with the allocated safety target.

Risk Assessment includes:

1) System definition,

2) Risk analysis,

3) Risk evaluation.

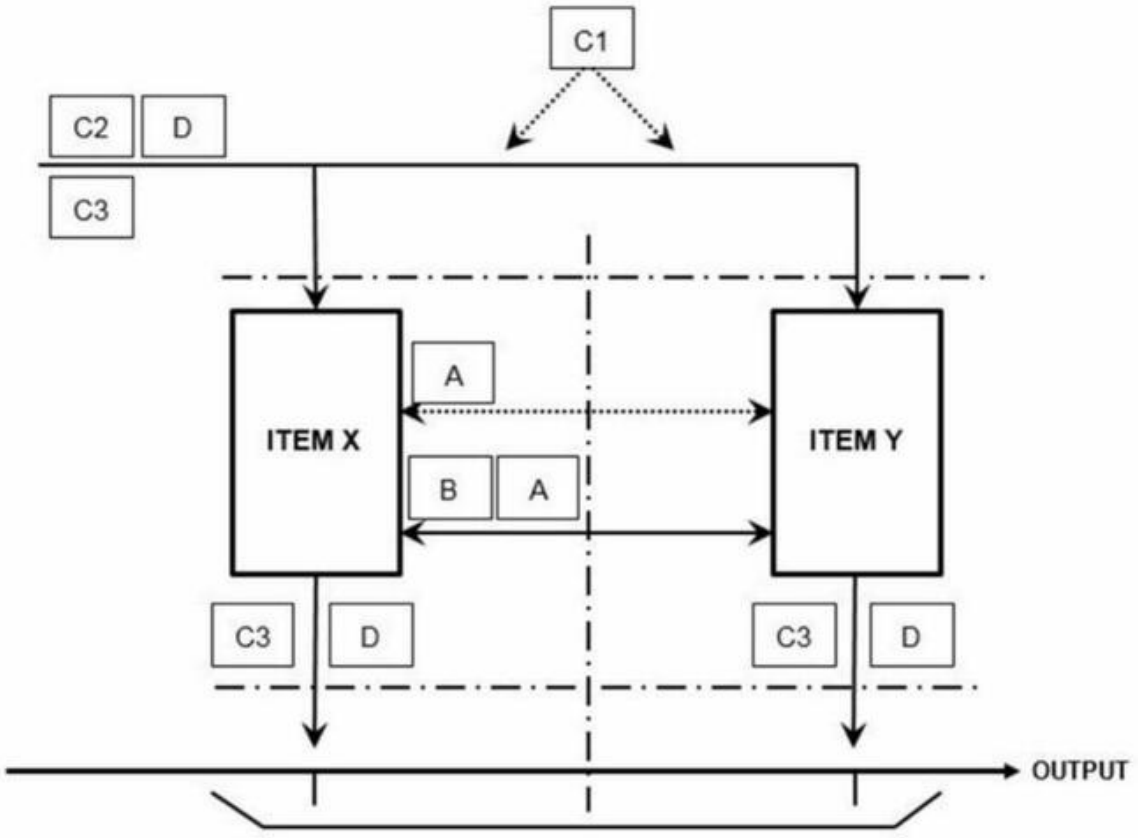
NOTE Typically within the Hazard Control for an electronic system the techniques and measures implemented will address reduction of risk on the basis of occurrence frequencies and not of reduction of consequences.

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Legend:

- Intentional connection
- Non-intentional connection (possibly caused by a fault)
- Independence (if specified measures are met to avoid non-intentional influences and connections)
- Front contact (normally-open contact)
- Two front contacts (used symbolically as an AND for two independent non-restrictive activities)
- A — Physical internal influence (non-intentional)
- B — Functional internal influence (non-intentional, using intentional connection)
- C1 — External environmental influence (EMI, ...) (non-intentional)
- C2 — External influence by power supply (non-intentional, using intentional connection)
- C3 — External influence across input/output (process working voltages, EMI-induced voltages) (non-intentional, using intentional connection)
- D — Functional external influence (non-intentional, using external connection)



"AND" condition for the non-restrictive state of the output

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